

This is sample data provided for CALIBER purposes only	ONE POINT LESSON (OPL)	OPL No: OPL-GA-1201A-07 Sheet 1 of 1
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OPL Title	Vibration Trend Monitoring & Alarm Response (VSHH-1201)	Discipline	Instrument
Equipment	GA-1201A - HEXANE FEED PUMP	Area / Unit	1200 - HEXANE FEED SYSTEM
Related Interlock	SEQ-1201 (SIL 1)	P&ID; Ref	TJC-LLD-PID-1201

Classification:	☐ Basic Knowledge	☒ Improvement	☐ Trouble Case	Aspect: P-Q-C-D-S-M-E
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1. PURPOSE / OBJECTIVE

This One Point Lesson explains **vibration trend monitoring & alarm response (vshh-1201)** for GA-1201A (HEXANE FEED PUMP) in the HEXANE FEED SYSTEM. A rising vibration trend predicts bearing failure days in advance and lets you plan a repair before a trip. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **n-Hexane** at up to 16 barg / 80 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Record VT-1201 on the daily PdM route in mm/s RMS	Loop isolated / under permit
2	Note alarm at 4.5 mm/s and trip at 7.1 mm/s (VSHH-1201)	Reference standard traceable
3	A rise greater than 1 mm/s within a week = plan a bearing inspection	As-found & as-left recorded
4	Correlate the vibration rise with TI-1201 bearing temperature	Trip / control action verified
5	Escalate to reliability engineer if the 1x running-speed component dominates	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Scheduled proof test of low-low suction pressure trip	Switch set-point drift found +0.08 barg during proof test	Recalibrated PSL-1201 to 0.5 barg trip, function tested to DCS
PDI-1201 reading erratic, seal-flush low alarm nuisance-tripping	Impulse lines fouled with hexane residue/polymer fines	Cleaned impulse lines and re-zeroed PDI-1201 transmitter
Discharge flow control hunting, FIC-1201 oscillating	Loop poorly tuned after transmitter re-range	Re-ranged FT-1201 and retuned FIC-1201 PID parameters

6. KEY LEARNING POINTS

- A rising vibration trend predicts bearing failure days in advance and lets you plan a repair before a trip.
- Respect the interlock SEQ-1201: e.g. PSL-1201 trips at < 0.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-1201 before working on GA-1201A.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
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